

International Institute of Information Technology, Hyderabad
CS6.302: Software System Development — Mid Semester Examination
Question Paper with Brief Solutions

Q1: Fill in the Blanks (20 Marks)

- a. **Question:** An async function implicitly wraps any non-Promise return value inside a _____ Promise, but if an uncaught error is thrown inside its execution body, it automatically returns a _____ Promise.
Answer: resolved (or fulfilled) ; rejected
- b. **Question:** While both CTEs and subqueries act as temporary result sets, a major advantage of a CTE over a standard inline subquery within a single execution statement is that a CTE can be referenced multiple times without repeating its query definition. [True/False]
Answer: True
- c. **Question:** If a user clicks the "Cancel" button or presses the Escape key on a prompt() dialog, the method returns _____ whereas dismissing an alert() dialog returns _____.
Answer: null ; undefined
- d. **Question:** In the browser environment, functions like alert() and confirm() belong to the _____ object, whereas DOM methods such as getElementById() and createElement() are invoked on the _____ object.
Answer: window ; document
- e. **Question:** To provide pre-filled text in the input field of a prompt box, you pass it as the _____ parameter in prompt method.
Answer: second (or default)
- f. **Question:** To completely remove an existing element from the DOM using its reference, you invoke the method element._____.
Answer: remove
- g. **Question:** Unlike updateOne(), which updates specific fields using operators like \$set, MongoDB provides the method db.users._____({_id: id}, newDoc) when the intention is to overwrite the entire document.
Answer: replaceOne
- h. **Question:** To retrieve only those users who have a status of "Active" or an age of 25, fill in the appropriate logical operator below: db.users.find({ _____ : [{ status: "Active" }, { age: 25 }] });
Answer: \$or
- i. **Question:** A newly initialized JavaScript Promise instance begins in the _____ state, and once it transitions to either fulfilled or rejected, its state becomes immutable; while synchronous exceptions are caught by try...catch, handling rejections from an un-awaited Promise chain requires appending the _____ method.
Answer: pending ; .catch()
- j. **Question:** In React class components, the render() method conceptually behaves like an idempotent function because it must produce the exact same UI output for the same input without side effects. When returning UI using JSX syntax, modern browsers cannot execute it natively, so a compiler such as _____ is required to transpile the code into standard JavaScript calls to _____.
Answer: Babel ; React.createElement

k. Question: In a JavaScript switch (val) statement, case clauses are evaluated lazily and compared against val using the _____ comparison operator; consequently, passing new Boolean(false) into a switch matching case false: will evaluate to _____.

Answer: strict equality (===) ; false (no match)

l. Question: An INOUT parameter can receive a constant literal directly from the caller, but the stored procedure is restricted from assigning or modifying its value internally. [True/False]

Answer: False

m. Question: To select an HTML element with an ID of header, CSS uses the _____ symbol before the identifier name.

Answer: # (hash)

n. Question: To fetch just a single document representing the record for "Alice" from the collection, rather than returning a cursor of multiple matching documents with find(), the appropriate method to use below is: db.users._____({ name: "Alice" });

Answer: findOne

Q2: Short Conceptual Questions (15 Marks)

a. Why do we need this function in JavaScript? Can we live without this?

Answer: The `this` keyword provides dynamic execution context, allowing methods to operate on the specific object that invoked them and enabling shared logic across instances without memory duplication. While modern functional programming with closures can largely avoid `this`, we cannot completely live without it because JavaScript's prototype chain, ES6 classes, and standard browser DOM event models fundamentally depend on it.

b. Explain difference between variable definitions `var`, `let`, `const` in JavaScript

Answer: `var` is function-scoped and hoisted with an initial value of `undefined`. In contrast, `let` and `const` are block-scoped and hoisted into the Temporal Dead Zone (TDZ), throwing an error if accessed before declaration. Furthermore, `var` and `let` allow reassignment, while `const` enforces read-only binding and disallows variable reassignment.

c. What is Map-Reduce. Illustrate with a very small example.

Answer: Map-Reduce is a programming model for data processing where "Map" transforms each input item independently into an intermediate format, and "Reduce" aggregates and combines the mapped results into a single value.

Example:

```
const nums = [1, 2, 3];
const squares = nums.map(n => n * n);           // Map: [1, 4, 9]
const sum = squares.reduce((acc, val) => acc + val, 0); // Reduce: 14
```

d. What are HTML Classes? Illustrate with one example.

Answer: HTML classes are non-unique attributes applied to elements to assign them to a shared category for CSS styling and DOM selection. Unlike IDs, multiple elements can share the same class name.

Example:

```
<p class="highlight">First paragraph</p>
<p class="highlight">Second paragraph</p>
```

e. DOM is a platform specific and language dependent programming interface. Argue if this statement is true or false with a brief reason.

Answer: False. The W3C specification explicitly defines the Document Object Model (DOM) as a platform-neutral and language-independent interface. Although most commonly used with JavaScript in web browsers, it can be implemented and accessed using various programming languages including Python, Java, and C++.

Q3: JavaScript Scripts (15 Marks)

(a) Script to illustrate an infinite loop: [2 Marks]

```
while (true) {
    // Executes indefinitely
}
```

(b) Script to illustrate push, pop, shift, unshift, and splice on an array: [5 Marks]

```
let arr = [20, 30];
arr.push(40);           // [20, 30, 40] - adds to end
arr.unshift(10);        // [10, 20, 30, 40] - adds to start
```

```

let popped = arr.pop(); // returns 40 -> [10, 20, 30]
let shifted = arr.shift();// returns 10 -> [20, 30]
arr.splice(1, 0, 25);    // [20, 25, 30] - inserts 25 at index 1

```

(c) Script to illustrate public, private, static, and privileged scope of variables: [4 Marks]

```

class Account {
  accountType = "Savings";      // Public field
  #balance = 1000;              // Private field (# prefix)
  static bank = "IIIT Bank";    // Static field

  constructor(initialPin) {
    let pin = initialPin;       // Closure-scoped private variable
    // Privileged method (accesses closure and private state)
    this.verifyAndGetBalance = function(inputPin) {
      return inputPin === pin ? this.#balance : "Unauthorized";
    };
  }
}

```

(d) Script to illustrate async/await with Promise: [3 Marks]

```

function fetchData() {
  return new Promise((resolve) => setTimeout(() => resolve("Success"), 500));
}

async function run() {
  try {
    const result = await fetchData();
    console.log(result);
  } catch (error) {
    console.error(error);
  }
}

run();

```

(e) Script to illustrate multiline alert: [1 Mark]

```

alert("Line 1: Roll number entered.\nLine 2: Proceed to exam.");

```